



L10 MOC IT Security Guidance

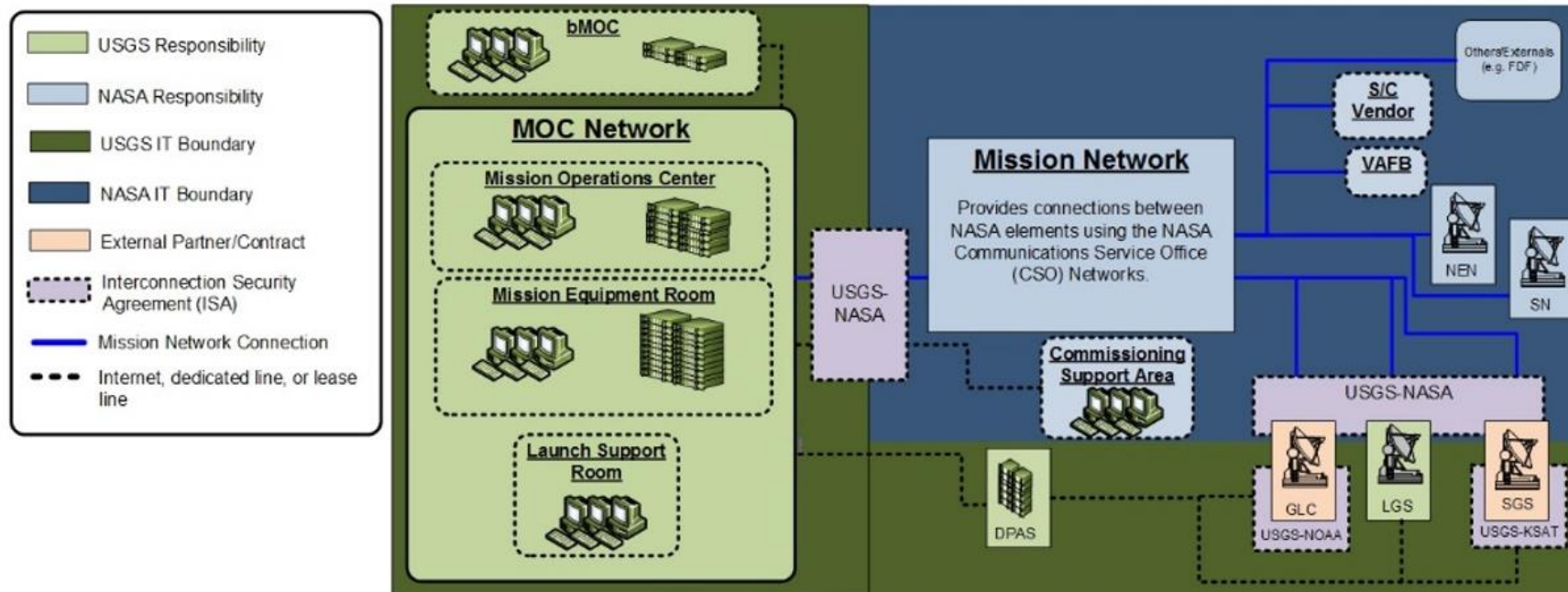
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- ◆ **Overview and Outline**
- ◆ **Security Documentation**
- ◆ **MOC Boundaries and Authorizations**
- ◆ **Directives and Mandates**
- ◆ **Data Protection**
- ◆ **Existing Missions IT Security Overview**

- ◆ **Assessment and Authorization (A&A)**
 - ◆ NIST RMF Framework
 - ◆ Ongoing FedRamp Compliance
 - ◆ Federal Information Security Modernization Act (FISMA)
 - ◆ Risk Identification or Plan of Action and Milestones (POAM)
 - ◆ Continuous Authorization to Operate (ATO) – Annual Renewal
 - ◆ Evaluation of Controls (Self-Assessment)
- ◆ **IT systems will leverage modern Security architecture and controls**
 - ◆ I.E. Zero Trust Architecture (ZTA)
- ◆ **IT systems will require Continuous Diagnostics and Mitigation tools**
 - ◆ Hardware Asset Management (HWAM)
 - ◆ Security Information and Event Management (SIEM)
 - ◆ Endpoint Protection
 - ◆ Vulnerability Management
 - In accordance with mandated identification and mitigation timelines

- ♦ **The deliverables of this study do not require any security documents to be completed**
- ♦ **The following documents are typically required as part of the ground systems security plan**
 - ♦ FIPS-199
 - ♦ Interface Control Document (ICD)
 - ♦ Interconnection Security Agreement (ISA)
 - ♦ Memorandum of Understanding/Agreement (MOU/A)
 - ♦ Information System Contingency Plan (ISCP)
 - ♦ Business Impact Analysis (BIA)
 - ♦ System Security Plan (SSP)
 - ♦ Continuity Of Operations Plan (COOP)
 - ♦ Privacy Impact Assessment (PIA)
 - ♦ Etc
- ♦ **For L10, it's expected that the MOC will provide content that will be used to construct these documents or use the documents in the overall security compliance**

- ♦ **IT security boundaries for the L10 MOC system have not been defined at this time**
 - ♦ NASA boundaries consist of mission networks that provide connection to NASA elements using NASA communications service office (CSO) networks
 - Connect to NSN stations, FDF, Spacecraft Vendor facility, launch facility, Space Network, etc.
 - ♦ USGS boundaries consist of the MOC, DPAS and GN systems
 - ♦ Will require M-22-09 Zero Trust Architecture
- ♦ **Conceptual diagram that was used for Landsat 9 during development:**



- ◆ **Adapt and implement dynamic Directives and Mandates as required. Examples include:**
- ◆ **M-22-09**
 - ◆ ZTA
 - ◆ Multi-Factor Authentication
 - ◆ Data in Transit
 - ◆ Data at Rest
- ◆ **M-21-31**
 - ◆ Logging capabilities and retention
- ◆ **BOD 22-01**
 - ◆ Known Exploited Vulnerabilities
- ◆ **BOD 23-01**
 - ◆ Visibility and Vulnerability Detection

◆ Dissemination Controls

- ◆ Data handling in accordance with labeling and privacy handling (CUI/PII/ITAR)

◆ TLM and CMD Data Protections

- ◆ Encrypted commanding (encryption takes place in MOC before being sent to stations)
- ◆ Encryption standard is notionally AES-256 or better
- ◆ Some S/C information (i.e. – S/C ID) is considered protected information, but much of this will be handled in T&C data formatting/definitions instead of specific transport protections
- ◆ TLM D/L is unencrypted, but will need to be protected in transit between stations and MOC (SSL/TSL tunnel or better)
- ◆ Potential for requirement for TLM D/L encryption

- ♦ **Landsat IT Security implementation and review is a joint USGS and NASA endeavor**
 - ♦ A joint Agency Security Working Group will be put in place and will continue to meet through the life of the mission
 - ♦ All government security standards (e.g., FISMA 2002) and industry practices will be implemented on all IT systems, including MOC systems
 - ♦ NIST 800-53 Rev5 (or latest) Security Controls will be the basis of annual compliance and adherence to federal security standards
 - ♦ Security standards for the MOC at GSFC will be a high watermark of USGS and NASA IT Security requirements
 - Security Standards for the NASA Exclusive Elements (CSO, FDF, CARA, etc...) comply with there applicable NASA IT Security Requirements
- ♦ **Landsat 9 Rated as Moderate Impact System for Security Planning and Implementation activities by both Agencies**
 - ♦ Security level not defined for Landsat 10 at this time
 - ♦ A FIPS 199 assessment will be performed as the mission progresses through phase A

- ◆ **USGS encourages the most efficient methods for implementing security for the Landsat 10 MOC**
 - ◆ Any feedback is greatly appreciated